

Lab-8 SSL/TSL Certificate

Task-1: . SSL/TLS Certificate Validation

Tool: OpenSSL, Web Browsers

Practical:

Check the SSL/TLS certificate of a website.

Inspect the expiration date, issuer, and encryption details.

Validate a certificate chain.

- Website Tested: <https://example.com/>
- Issuer: DigiCert Inc
- Validity: From **15 Jan 2025 00:00:00 GMT**
To **15 Jan 2026 23:59:59 GMT**
- Status: **Valid**

```

1:hackergkai - x
o echo | openssl s_client -connect example.com:443 2>/dev/null | openssl x509 -noout -text
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            0a:08:15:ba:fa:48:b0:b7:f9:7a:60:4f:06:ac:1e:9a
        Signature Algorithm: ecdsa-with-SHA384
        Issuer: C=us, O=DigiCert Inc, CN=DigiCert Global G3 TLS ECC SHA384 2020 CA1
        Validity
            Not Before: Jan 15 00:00:00 2025 GMT
            Not After : Jan 15 23:59:59 2026 GMT
        Subject: C=us, ST=California, L=Los Angeles, O=Internet Corporation for Assigned Names and Numbers, CN=*.example.com
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
            Public-Key: (256 bit)
            pub:
                04:9a:48:97:84:7d:a1:6e:08:c9:6a:14:a0:c8:38:
                80:ac:00:00:07:fa:97:0a:13:80:a2:0e:87:92:
                57:a7:08:f9:3c:1a:3d:0c:18:47:03:37:63:0a:13:8a:
                43:02:71:74:bd:75:0f:4e:07:05:ce:02:2b:a0:e4:
                6a:72:c2:cf:ae
            ASN1 OID: prime256v1
            NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                8a:22:eb:fc:a0:07:37:00:fb:40:21:39:76:9a:a1:67:0e:10:a8
            X509v3 Subject Key Identifier:
                f0:c1:64:32:00:ec:04:c7:e8:0f:cd:00:60:19:12:59:d1:0e:72:ed
            X509v3 Subject Alternative Name:
                DNS=*.example.com, DNS=example.com
            X509v3 Certificate Policies:
                Policy: 2.5.149.1.2.2
                CPS: http://www.digicert.com/CPG
            X509v3 Key Usage: critical
                Digital Signature, Key Agreement
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
            X509v3 CRL Distribution Points:
                Full Name:
                    URI:http://crl3.digicert.com/DigiCertGlobalG3TLSECCSHA38402020CA1-2.crl
                Full Name:
                    URI:http://crl4.digicert.com/DigiCertGlobalG3TLSECCSHA38402020CA1-2.crl

```

[illegible]

Task-2: Web-Based SSL/TLS Certificate Testing:

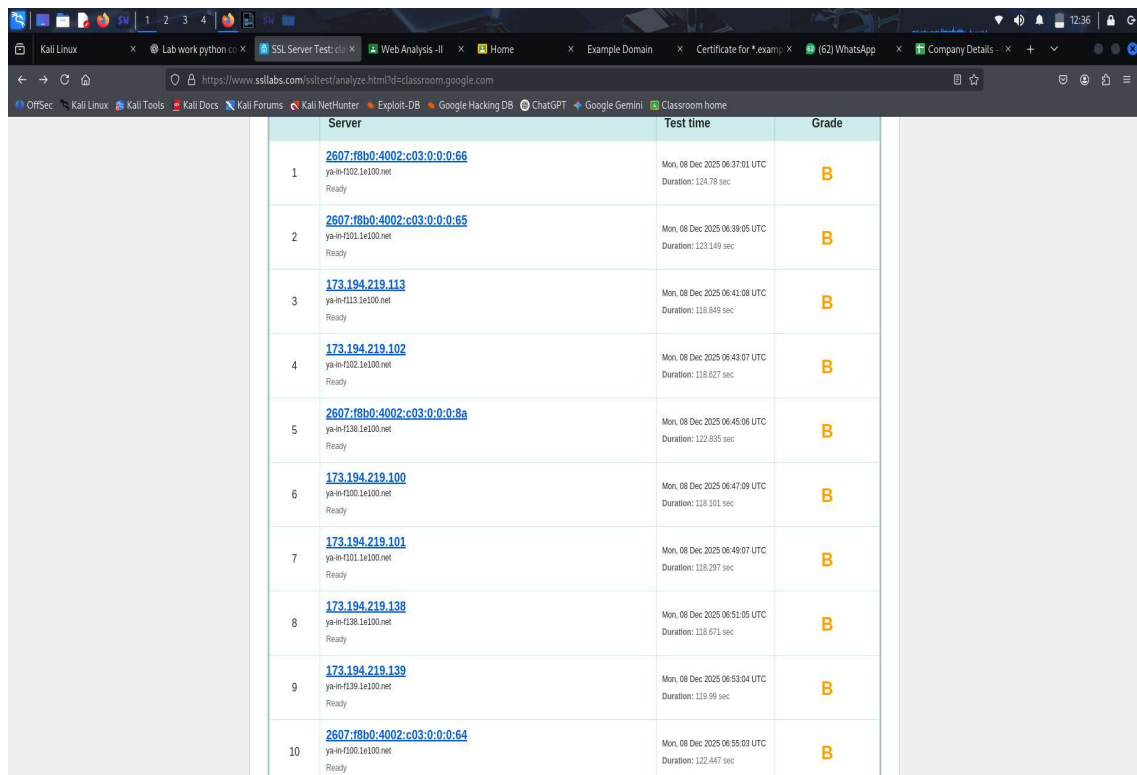
Tool: SSL Labs

(www.ssllabs.com/ssltest/)

Practical: Analyze the SSL/TLS security of a website.

Identify vulnerabilities like weak ciphers or expired certificates.

- For the scanning in ssl labs website, i have used this url:
<https://classroom.google.com/> of classroom home page for scanning purpose.
- As per the lab scan result, the Grade for this website was found as “B” from 10 different servers which have scanned for this website.



Server	Test time	Grade
1 2607:f8b0:4002:c03:0:0:0:66 ya-h-f102.1e100.net Ready	Mon, 08 Dec 2025 06:37:01 UTC Duration: 124.78 sec	B
2 2607:f8b0:4002:c03:0:0:0:65 ya-h-f101.1e100.net Ready	Mon, 08 Dec 2025 06:39:05 UTC Duration: 123.149 sec	B
3 173.194.219.113 ya-h-f113.1e100.net Ready	Mon, 08 Dec 2025 06:41:08 UTC Duration: 118.849 sec	B
4 173.194.219.102 ya-h-f102.1e100.net Ready	Mon, 08 Dec 2025 06:43:07 UTC Duration: 118.627 sec	B
5 2607:f8b0:4002:c03:0:0:0:8a ya-h-f138.1e100.net Ready	Mon, 08 Dec 2025 06:45:06 UTC Duration: 122.835 sec	B
6 173.194.219.100 ya-h-f100.1e100.net Ready	Mon, 08 Dec 2025 06:47:09 UTC Duration: 118.101 sec	B
7 173.194.219.101 ya-h-f101.1e100.net Ready	Mon, 08 Dec 2025 06:49:07 UTC Duration: 118.297 sec	B
8 173.194.219.138 ya-h-f138.1e100.net Ready	Mon, 08 Dec 2025 06:51:05 UTC Duration: 118.671 sec	B
9 173.194.219.139 ya-h-f139.1e100.net Ready	Mon, 08 Dec 2025 06:53:04 UTC Duration: 119.99 sec	B
10 2607:f8b0:4002:c03:0:0:0:64 ya-h-f100.1e100.net Ready	Mon, 08 Dec 2025 06:55:03 UTC Duration: 122.447 sec	B

- The results showcase all details about the summary as:
 - This server supports TLS 1.0 and TLS 1.1. Grade capped to B.
 - This server supports TLS 1.3.DNS Certification Authority Authorization (CAA) Policy found for this domain.

III. The certificate is valid from 27 Oct 2025 08:35:33 UTC to 19 Jan 2026 08:35:32 UTC.

IV. Key - EC 256 bits, Issuer – WE2, Signature algorithm- SHA256 with ECDSA, Certificate Transparency – Yes.

V. Trusted – Yes, DNS CAA- Yes

- I have also attached a soft copy of the certificate issued from the ssl labs.

Task 3: Digital Certificate Creation & Management Tool: OpenSSL Practical: Generate a self-signed certificate using OpenSSL. Create a Certificate Signing Request (CSR). Sign a certificate using your own Certificate Authority (CA). Verify and inspect the certificate details. Export and import certificates in different formats (PEM, DER, PFX).

- For this lab, first of all i have created a ca.key private key.
- Then i have created a Self signed CA certificate, with the name as ca.crt, all the fields filled up in this manner
 - Country Name (2 letter code) [AU]:IN
 - State or Province Name (full name) [Some-State]:Gujarat
 - Locality Name []:Bharuch
 - Organization Name (eg, company) [Internet Widgits Pty Ltd]:My Test CA
 - Organizational Unit Name (eg, section) []:Test CA
 - Common Name []:test server
 - Email Address []:meet.2k21@gmail.com
- Here the ca.crt is the **root CA certificate** and is self-signed.

- Using openssl verify, the server certificate was successfully validated using the CA certificate, confirming a correct certificate chain in our mini PKI.

```

1: hacker@kali: ~
Data:
  Version: 3 (0x2)
  Serial Number:
    21:5e:01:a6:86:47:2b:a0:9e:81:ae:36:16:ad:c3:93:6d:46:04:5b
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C=IN, ST=Gujarat, L=Bharuch, O=My Test CA, OU=Test CA, CN=Test server, emailAddress=meet.2k21@gmail.com
  Validity
    Not Before: Dec  8 07:54:10 2025 GMT
    Not After : Dec  8 07:54:10 2026 GMT
  Subject: C=IN, ST=Gujarat, L=Vadodara, O=Test Labs, OU=Test, CN=My Testing Company, emailAddress=meet.2k21@gmail.com
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:9d:5c:72:a0:4d:2b:23:ad:54:26:78:53:d8:ed:
      f4:2f:af:9:ae:ee:23:41:3c:a5:50:83:ca:7a:cd:
      15:18:df:52:b5:40:2d:8f:a0:88:5b:cd:5b:b4:0b:
      ef:a4:fc:f1:61:3b:7d:62:c2:cd:24:90:9b:69:b5:ed:
      4d:36:4e:fb:04:cb:e7:ae:72:07:b2:e7:04:2f:cd:
      bf:ed:fc:e7:57:22:5d:0e:34:70:3f:b0:9b:c1:03:
      3b:b6:bf:c9:4e:8b:41:29:c1:50:f1:7b:f5:a2:bf:
      81:d3:c6:58:fb:d2:b4:f2:eb:75:77:a9:4f:b9:74:
      c8:ae:b2:e1:25:c1:ef:cf:90:80:0f:3b:df:fd:76:
      8e:79:58:38:4e:19:df:8c:f8:12:39:72:5b:7e:ae:
      30:d6:50:58:a8:bd:6f:a2:ee:14:bb:10:66:4b:8e:
      fa:e7:f0:69:27:a0:d4:91:cb:39:7a:48:93:2b:83:
      7f:f9:b2:35:c7:6a:e5:69:f8:6a:aa:ff:eb:bca:8:
      18:4a:1b:cb:9f:42:8d:52:14:e7:f5:0b:d7:32:4d:
      28:78:83:94:93:98:77:3b:77:b3:12:0b:7a:73:c8:
      b5:46:e2:85:da:fe:b3:cd:11:24:3f:be:e0:be:8a:
      8b:54:aa:97:57:81:ca:5d:2e:af:da:5c:dc:42:5a:
      68:3f
    Exponent: 65537 (0x10001)
  X509v3 extensions:
    X509v3 Subject Key Identifier:
      EA:7C:81:28:D4:73:D8:26:00:F1:D0:16:3F:85:6D:C6:63:D0:48:B2
    X509v3 Authority Key Identifier:
      ED:E8:6E:62:AE:19:D6:F4:F4:42:C0:2C:37:4A:19:46:E3:3E:2E:76
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    25:83:29:82:4b:26:41:c6:74:f3:18:dd:da:54:ab:96:33:20:
    64:35:10:70:28:45:0d:b6:2b:c1:0b:61:29:38:39:40:b6:53:
    2e:a6:a1:2d:d8:76:01:1b:7f:35:fe:8a:73:69:bc:bb:f4:2f:

```

- This image is the output for the verification of the certificate which was run in terminal by using ssl verification is successful against CA.